Real-Time Smart Energy Meter with Sensor-Based Monitoring and In-Device Payment Capability

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***Abstract*—This paper presents an IoT based smart energy metering system designed for residential and commercial use. It leverages an ESP32 microcontroller paired with ZMPT101B and ZMCT103C sensors, the system measures voltage and current in real time, calculates power consumption, and reports data via MQTT to a HiveMQ cloud broker. A Node RED dashboard provides live visualizations, slab based billing, and UPI QR code payments. A novel voltage comparison algorithm detects localized outages, while Twilio API triggers SMS alerts for over /under voltage or abnormal usage. Experiment results show that ±5% accuracy as compared with standard meters and reliable operation across multiple networked nodes. The result is a scalable, cost effective, user friendly alternative enhancing modern smart grid infrastructures.**

***Index Terms*—Smart energy meter, ESP32, IoT, MQTT, QR code, Node-RED, Twilio, real-time monitoring.**

I. Introduction

The rapid evolution of urbanization and the proliferation of digital devices have led to a surge in global energy demand, posing significant challenges to legacy metering infrastructure. Traditional energy meters, which typically rely on manual reading and fixed-rate billing, lack the responsiveness and transparency required for effective energy management in smart grid environments. Furthermore, they fail to provide real time insights, adaptive pricing mechanisms, or automated fault detection all of which are critical for modern energy systems. In response to these limitations, recent advances in the Internet of Things (IoT) have enabled the development of smart metering systems capable of delivering real-time energy monitoring, dynamic pricing, and user-centric billing mechanisms. They not only record consumption but also facilitate time-sensitive billing, remote control, and in-device payments, enabling advanced demand-side management and energy market participation. These developments address global sustainability goals by enhancing operational efficiency, reducing losses, and empowering consumers.

The revolution of smart grids has been driven by layered enhancements in smart metering, communication infrastructure, and data-driven intelligence. Early innovations included IoT-enhanced power meters for routing energy based on quality [1], secure multiparty protocols for privacy-preserving aggregation [2], and distributed, self-configuring metering networks for real-time demand response [3]. Further progression involved embedding business-intelligence capabilities into smart meter systems [4], and expanding grid resilience via communication technologies tailored to renewable energy flows [5] [7]. Comprehensive analyses of big-data applications in these systems underscored scalable analytics and cybersecurity considerations [6]. The integration of IoT marked a turning point: Saleem et al. provided architectural blueprints-HAN/NAN/WAN layered systems—with prototypes for smart grids [8], while mapped the landscape of IoT-powered energy management and system awareness [9]. Prototypical systems followed: an Arduino/GSM meter for remote monitoring [12], cloud-integrated real-time meters [16], and modular meters for diverse load scenarios [17]. Platforms using NB IoT [37], embedded anomaly detection in EV contexts [21], and IoT-cloud systems for home energy control [32] [33] emerged, demonstrating both breadth and adaptability. Advanced analytics and security matured in parallel: Maitra et al. demonstrated convolutional-autoencoder-based anomaly detection with dynamic thresholds [22], while smart-meter analytics frameworks transitioned from descriptive to predictive and prescriptive paradigms [23]. Robust mechanisms for theft detection-via privacy-preserving dynamic billing in AMI systems [24], IoT-based theft frameworks [25], and AMI-rooted localization/detection methodologies [26], [27] were complemented by anti-tampering and pattern-based anomaly models [28], [29].

Power quality and dynamic billing became mainstream: PQ-smart meters were deployed, capable of penalizing harmonic and reactive distortion [15]; In [16] and [17], introduced modular IoT-based power meters across residential and industrial domains; Modern meters against analog counterparts across dynamic pricing, accuracy, and consumer behavior metrics were evaluated [19]; while wireless PQ meters balanced affordability and functionality [20]. Technology adoption across North America and Africa, with AI and blockchain implementations for socio-economic upliftment [30]; Fuzzy-logic EMS delivered up to 34% peak-load reduction [31]; ML-centric real-time optimization in smart homes [34]; Time-series forecasting in cloud-connected meters [36]; NB IoT with cloud backends [37]; Institutional energy systems [35].

This paper proposes a novel IoT-based Smart Energy Metering System built around the ESP32 microcontroller, incorporating multiple smart features that extend beyond basic monitoring. The system integrates ZMPT101B and ZMCT103C sensors for high-resolution voltage and current measurements, processes data using the EmonLib library, and communicates over MQTT protocol to a cloud-hosted HiveMQ broker. A custom-designed Node-RED dashboard visualizes energy metrics and supports slab-based pricing and dynamic billing in real time.

A key innovation introduced in this work is the use of a neighboring meter voltage comparison mechanism, which enables the system to detect localized outages or discrepancies in power supply, a feature seldom addressed in existing literature. Additionally, the integration of Twilio-based SMS alerts ensures users are promptly notified of abnormal conditions such as overvoltage, undervoltage, and threshold violations. The system also supports QR code-based payment via UPI and digital wallets, transforming the energy meter into a complete end-to end billing and payment platform. The proposed system was evaluated under simulated residential and commercial conditions, showing high reliability, low latency communication, and end-user accessibility across platforms. By consolidating sensing, communication, billing, and payment within a modular architecture, this work contributes a scalable and cost-effective solution for smart grid deployments. The growing need for transparent and efficient energy consumption monitoring has catalyzed the development of smart metering systems that leverage Internet of Things (IoT) technologies. Several studies have explored energy meters integrated with wireless communication, remote monitoring, and basic alert mechanisms. However, many of these systems fall short in terms of adaptability, dynamic pricing, fault resilience, and user-centric interaction. In contrast, the system proposed in this paper combines hardware-level energy sensing, dynamic billing, visual analytics via Node-RED, real-time anomaly detection, neighbor-aware. outage diagnostics, and integrated digital payment capabilities. This end-to-end architecture addresses key shortcomings in existing research and presents a practical, deployable model for next-generation smart grid metering solutions.

II. System Design and Architecture

Karnataka is actively transitioning to smart metering, yet the rollout continues to face significant hurdles. Although BESCOM has begun installing smart meters in new and temporary connections at a cost of approximately 4,998 per unit plus a maintenance fee of 75 / month, this rollout remains partial and existing consumers continue to pay the service cost. Smart meters are currently mandatory only for new connections, with state authorities clarifying that replacements of existing functioning meters are optional, as tested in a PIL in the Karnataka High Court. In addition, questions about transparency, procurement costs, and value for money persist, as smart meter prices in Karnataka exceed 5,000, much higher than those in other states where similar meters cost closer to 900. Despite central schemes like RDSS offering subsidies, Karnataka hasn't fully opted in, citing concerns over financial viability, substantial pending dues, and the need for infrastructure upgrades instead of meter replacement. This staggered rollout and cost burden on consumers highlight persistent interoperability issues and limited backend integration, as backend server links are still being activated for only a fraction of installed units.

Hence, proposed IoT-based Smart Energy Metering System directly addresses these regional challenges. It combines low-cost hardware (ESP32 with ZMPT101B/ZMCT103C), efficient MQTT communication, and cloud-based dashboards to achieve ±5% accuracy comparable to traditional meters. It also features dynamic slab billing, QR-coded UPI payments, outage detection via neighboring meter comparison, and automated SMS alerts. This approach reduces consumer costs, enhances backend compatibility, and boosts grid transparency and resilience. Given Karnataka's phased rollout, infrastructure constraints, and public concern, the system offers a practical, scalable, and secure solution ideally suited for both residential and commercial contexts-supporting a smarter, more responsive, and equitable energy future in the state.

A. Hardware Architecture

At the core of the system is the ESP32 microcontroller, selected for its dual-core performance, built-in Wi-Fi and Bluetooth, and compatibility with high-frequency analog sensing. This integrated system employs the ZMPT101B module for accurate AC voltage sensing, delivering an isolated signal proportional to the grid voltage. For current measurement, it utilizes the ZMCT103C sensor, capable of precisely monitoring currents up to 5A, suitable for both residential and commercial applications . A secondary ESP32 module, configured with a digital input such as a push button or simulated grid voltage input, functions as a neighboring node to cross-verify power availability and assist in diagnosing outages. An optional relay control module enables the system to disconnect loads during sustained overvoltage, undervoltage, or when critical energy limits are breached, enhancing safety and reliability. To ensure stable and noise-immune operation over extended periods, dedicated power supply modules regulate voltage for the sensors and the ESP32 board. All sensors are connected to the ESP32's ADC channels with proper isolation, filtering, and calibration to prevent false readings and interference. Th schematic diagram of smart meter and its neighbouring meter is shown in Fig. 1 and 2 respectively.

B. Software Architecture

The ESP32 firmware orchestrates comprehensive energy monitoring by managing sensor data acquisition, signal conditioning, and executing real-time energy computations, all while handling MQTT communication to a cloud-hosted HiveMQ broker with minimal latency and high reliability. Leveraging the EmonLib library (adapted for ESP32), it computes root mean square (RMS) values for voltage and current, estimates apparent and real power, calculates power factor, and tracks cumulative energy consumption in kilowatt-hours

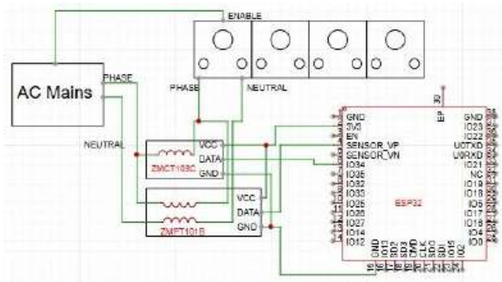


Fig. 1: Schematic diagram of smart meter

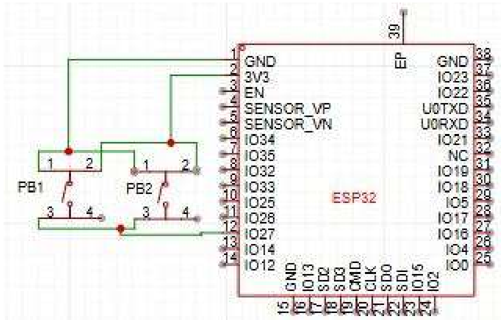


Fig. 2: Schematic diagram of neighbour meter

(kWh). Published data are relayed over MQTT to HiveMQ Cloud, where Node-RED subscribes to topics and drives a user-facing dashboard providing real-time graphs of voltage, current, power, and energy usage, slab-based billing and pricing breakdowns, system health warnings, and even dynamic QR code generation for UPI/digital wallet payments based on usage.The block diagram of the proposed system architecture is shown in Fig. 3.

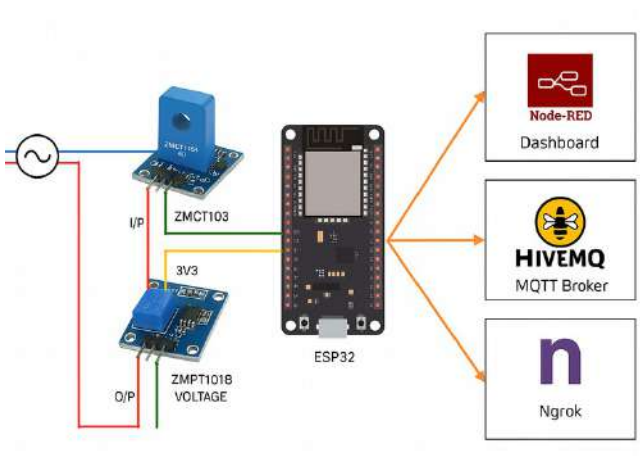


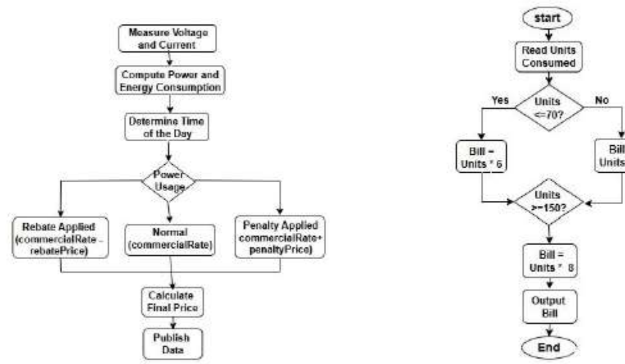
Fig. 3: Block diagram of the proposed system architecture

C. Communication and Integration Flow

The system follows a publish-subscribe model via MQTT, allowing bidirectional interaction. Twilio's SMS API is triggered from the ESP32 or from Node-RED in critical scenarios, including:

- Overvoltage or undervoltage detection
- Exceeding slab threshold limits
- Localized power outages (identified by mismatch between main and neighbouring meter states)

The flow diagrm is shown in Fig. 4a. Billing flow diagram is shown in Fig. 4b.



(a) (b)

Fig. 4: (4a).flow diagram of smart meter,(4b).flow diagram of billing

D. Key Innovations in Architecture

- Neighbouring Node Outage Detection: Unlike conventional meters, this system compares its voltage reading with that of a nearby ESP32 node to determine if an outage is local or grid-wide, enhancing diagnostic accuracy.
- Integrated QR Code Payment: The dashboard dynamically generates a QR code based on consumption and slab rates, enabling users to scan and pay through UPI or wallet apps with no additional payment gateway logic.
- Twilio-Driven Alert System: Unlike passive meters, this system actively notifies users of critical conditions via SMS, improving responsiveness and safety.
- Scalable MQTT-Based Framework: Supports expansion to multi-node or apartment-level deployments, with each node publishing to unique topics and a centralized dashboard for aggregated control.

III. IMPLEMENTATION AND METHODOLOGY

The implementation of the proposed Smart Energy Metering System follows a modular, multi-layered approach encompassing hardware interfacing, energy computation, real time communication, intelligent billing, fault detection, and user interaction. The overall methodology is structured across six core modules:

A. Sensor Interfacing and Data Acquisition

The ESP32 microcontroller serves as the central processing unit, acquiring analog signals from:

- ZMPT101B voltage sensor, interfaced via ADC for RMS voltage detection
- ZMCT103C current transformer, calibrated to measure RMS current with load variation

To enhance measurement stability and accuracy, input signals are filtered and sampled at high frequency using a timer-driven ADC routine. All calibration is done through empirical scaling with reference to a calibrated digital multimeter (DMM). Noise reduction is achieved using software-level low-pass filters. The EmonLib library is used for calculating:

- Voltage (Vrms), Current (Irms)
- Apparent Power: S = V × I
- Real Power: P = V × I × Power Factor
- Energy Consumption: E = P × t where t is time in hours
- Power Factor: computed from phase difference and load behavior.

Where V is the RMS voltage (from ZMPT101B)
I is the RMS current (from ZMCT103C)
Power Factor is calculated using the **EmonLib** library
Security and data efficiency are maintained through optimized topic hierarchies and payload structures.

B. Dynamic Billing and Slab-Based Pricing

The system implements slab-wise billing logic to accommodate both domestic and commercial users, following region-specific energy pricing policies. Energy data is sampled and accumulated over time to compute live billing.

1) Domestic Hourly Rate Calculation (Slab-Based): This is a custom slab model inspired by general domestic and commercial electricity pricing patterns in Karnataka. For domestic consumers, the electricity cost is calculated using a slab-based pricing system. Domestic Rate (Rs)

$$= \begin{cases} E \times \text{Rs.}3 & \text{if } E \leq 100 \text{ kWh}, \\ 100 \times \text{Rs.}3 & \\ +(E-100) \times \text{Rs.}4.50 & \text{if } 100 < E \leq 200 \text{ kWh}, \\ 100 \times \text{Rs.}3 + 100 \times \text{Rs.}4.50 & \\ +(E-200) \times \times \text{Rs.}6. & \text{if } E > 200 \text{ kWh}. \end{cases}$$

2) Commercial Hourly Rate Calculation : Commercial rates are typically higher. Commercial Rate (Rs)

$$= \begin{cases} E \times \text{Rs.}5. & \text{if } E \leq 100 \text{ kWh}, \\ 100 \times \text{Rs.}5. & \\ +(E-100) \times \text{Rs.}6.50 & \text{if } 100 < E \leq 200 \text{ kWh}, \\ 100 \times \text{Rs.}5 + 100 \times \text{Rs.}6.50 & \\ +(E-200) \times \times \text{Rs.}8. & \end{cases}$$

if **Ecurrent** $>$ **Ethreshold, then Notify the user by SMS** . where Ethreshold is the Maximum allowed units (e.g., 150 kWh) & Ecurrent is the Current energy consumed.

C. Voltage Quality and Neighbor-Aware Outage Detection

To ensure operational safety, the system continuously monitors grid voltage for abnormalities. Overvoltage: $>$ 260V Undervoltage: $<$ 110V When thresholds are exceeded, a fault flag is raised, and the relay module may disconnect the load. A second ESP32-based neighbouring meter is deployed to simulate or monitor an adjacent power line. Voltage comparison between meters determines: True outage (both meters show 0V) Local failure (primary 0V, neighbour active). This comparison enhances fault diagnosis without relying on cloud-based verification or external tools.

D. MQTT Communication Protocol

The system uses MQTT (Message Queuing Telemetry Transport), a lightweight protocol ideal for IoT deployments. The ESP32 publishes sensor and alert data to a HiveMQ broker over Wi-Fi. Payloads are JSON-formatted for ease of parsing by the Node RED dashboard or future integration with third-party platforms or utilities.

E. Real-Time Visualization and Payment Interface

The Node-RED dashboard subscribes to MQTT topics and visualizes:

- Live graphs of voltage, current, power, and energy
- Energy cost updated in real-time as per slab logic
- Status flags for faults, threshold breach, and outage detection

F. Automated Alert and Notification System

The system is integrated with the Twilio API to send SMS alerts to registered users. Alert types include:

- Voltage anomalies (over/under)
- Energy threshold exceeded
- Power outage detection (via neighbouring ESP32)

Twilio is triggered either directly via HTTP requests from the ESP32 or through Node-RED automation rules based on MQTT messages. The system also dynamically generates a QR code based on the user's accumulated bill. This QR is compatible with UPI and digital wallets, and can be scanned by users directly from the dashboard, allowing instant payment without banking portal redirection.

IV. RESULTS AND DISCUSSION

The proposed Smart Energy Metering System was prototyped and tested under simulated residential and commercial conditions using resistive and inductive loads ranging from 60W to 2000W. The performance was evaluated in terms of energy accuracy, slab pricing response, voltage anomaly detection, outage diagnosis, and payment system functionality.

A. Accuracy of Energy Measurements

To validate the accuracy of the sensing subsystem, readings from the ESP32-based meter were compared against a calibrated digital multimeter (UNI-T UT61E) and a commercial-grade energy meter (Secure Liberty 100).

$$Error(\%) = \frac{SmartMeter\ Reading - Ref.Reading}{Reference\ Reading} 100 \tag{1}$$

$$Accuracy(\%) = 100 - |Error(\%)| \tag{2}$$

Using equation 1 and equation 2, error and accuracy are calculated. The margin of error was consistently within ±5%,

TABLE I: Comparative analysis of electrical measurements

Parameter	Reference Reading	Smart Meter Reading	E(%)	A(%)
Voltage(Vrms)	217.49 V	219.88 V	+1.14	98.86
Current (Irms)	0.0877 A	0.0855 A	-2.47	97.53
Real Power	19.27 W	18.81 W	-2.43	97.57
Energy (1 hr)	1.392 Wh	1.362 Wh	-2.16	97.84

as shown in Table I. These results confirm that the system's energy measurements are within acceptable bounds for IoT-based non-industrial applications.

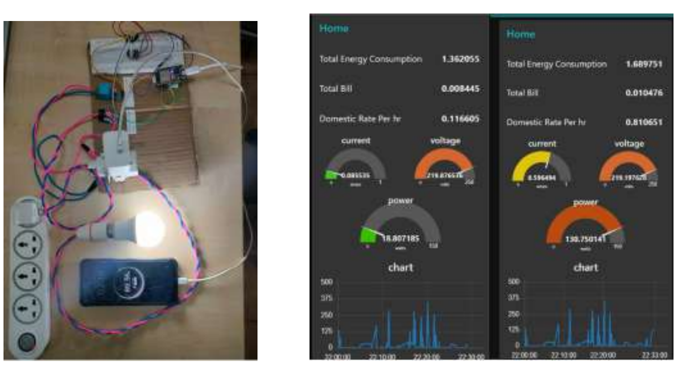
B. Real-Time Monitoring and Visualization

The system achieved consistent real-time data transmission to the HiveMQ broker, with an average publish latency of 178 ms and end-to-end dashboard update latency under 350 ms. The Node-RED dashboard rendered power, energy, and billing data without observable lag across mobile and desktop clients. The designed dashboard with menu bar with four options namly payment, previous data, domestic and industry use is shown in Fig. 5a and power distribution is displayed in Fig. 5b.

(a) (b)

Fig. 5: (5a). Node Red dashboard, (5b). Power distribution chart

The smartmeter is experimented with 9w LED bulb and 45W mobile phone charger is shown in Fig. 6a.The corresponding smart meter display is shown in Fig. 6b.



(a) (b)

Fig. 6: (6a).smart meter with two loads,(6b).reading with two loads

C. Slab-Based Billing Response

The slab-based billing mechanism was implemented in the system to accommodate both domestic and commercial energy pricing models. The system dynamically adjusted billing rates based on synthetic energy values input during testing. These test cases confirmed that

- The slab thresholds for domestic and commercial use triggered the correct pricing tier
- The calculated bill updated in real time on the Node RED dashboard
- Alerts were generated when simulated energy values exceeded predefined consumption thresholds

Additionally, exceeding user-defined thresholds (e.g., 150 kWh) triggered alerts via MQTT and SMS through Twilio. This closed loop behavior effectively demonstrated the pricing model's automation and user responsiveness.

D. Fault Detection and Voltage Anomalies

Voltage anomalies were simulated by introducing regulated voltage drops and surges using a variac. The system successfully detected:

- Over voltage (>260V): System-triggered MQTT + SMS alert
- Under voltage (<110V): Dashboard warning + relay disconnection (if enabled)

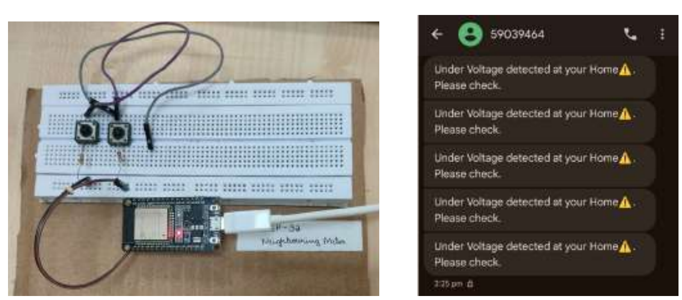
The reaction time between anomaly detection and alert dispatch was consistently below 600 ms.

E. Neighbour-Aware Outage Detection

A second ESP32 unit was deployed to act as a neighbouring meter with independent voltage input is shown in Fig. 7a. The message is shown in Fig. 7b. The main meter detected local outages by comparing its voltage status against that of the neighbouring node. Two scenarios were validated:

- True Outage: Both meters show 0V → Grid-level failure
- Local Failure: Primary shows 0V, neighbour shows

230V → Local disconnection (e.g., blown fuse) This neighbour-based diagnostic logic proved superior to single-node monitoring, providing an additional reliability layer not found in existing solutions.



(a) (b)

Fig. 7: (7a).Neighbour Meter,(7b).Alert message warning upon undervoltage detection

F. QR-Based UPI Payment

The Node-RED dashboard dynamically generated a UPI QR code based on the calculated bill amount. Users successfully completed test payments via Google Pay and PhonePe. Confirmation messages were simulated by webhook acknowledgment. This feature eliminates the need for external payment portals and can be extended to utility providers via API integration, which is shown in Fig. 8.

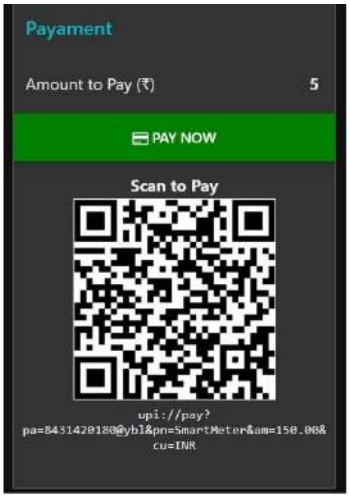
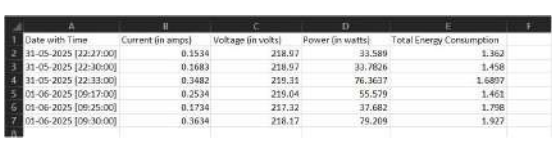


Fig. 8: QR code payment

G. Previous Data Logging and Retrieval

To enhance analytical capabilities and provide long-term visibility into energy usage, the system incorporates a historical data logging mechanism. All real-time energy parameters— such as voltage, current, power, and consumption—are persistently stored in a structured format (e.g., CSV) using Node-RED's file management tools. The stored data can be retrieved and visualized through a dedicated dashboard tab using searchable tables and time-series charts. This allows users to review past consumption patterns, compare billing cycles, and conduct usage-based diagnostics. The logging module is designed for low-overhead operation and can be extended to use local databases like SQLite or cloud storage platforms in future iterations. The previous date is stored in exel format. users can download and compare their usage chart and is shown Fig. 9.



	A	B	C	D	E	F
1	Date with Time	Current (in amps)	Voltage (in volts)	Power (in watts)	Total Energy Consumption	
2	31-05-2025 [22:27:00]	0.1534	218.97	33.589	1.362	
3	31-05-2025 [22:30:00]	0.1683	218.97	33.7826	1.458	
4	31-05-2025 [22:33:00]	0.3482	219.31	76.3637	1.6897	
5	01-06-2025 [09:17:00]	0.2534	219.04	55.579	1.461	
6	01-06-2025 [09:25:00]	0.1734	217.32	37.682	1.798	
7	01-06-2025 [09:30:00]	0.3634	218.17	79.209	1.927	

Fig. 9: Previous data

H. System Usability and Cross-Platform Support

The entire system was tested across:

- Browsers: Chrome, Edge, Firefox
- Devices: Windows laptops, Android smartphones
- Network types: Wi-Fi (2.4 GHz), Mobile Hotspot (5G LTE)

No performance degradation or communication failures were observed. The MQTT-based architecture and Node-RED dashboard proved resilient, fast, and user-friendly across platforms.

I. Comparative Evaluation with Existing Systems

Compared to prior works , the proposed system uniquely combines:

- Real-time sensing with dynamic pricing
- Twilio-integrated SMS alerting
- Neighbouring node voltage cross-check
- UPI-based QR payment integration

This integrated end-to-end feature set is largely absent in most peer systems, validating the system's novelty and deployability in smart grid infrastructure.

The proposed system demonstrates a viable path toward scalable, secure, and intelligent energy management solutions suitable for deployment in modern smart grid infrastructures.

V. Conclusion

This paper presented the design and implementation of an IoT based Smart Energy Metering System that integrates real-time monitoring, dynamic pricing, fault detection, and seamless digital payment. Leveraging the ESP32 microcontroller, ZMPT101B and ZMCT103C sensors, MQTT protocol, and a Node-RED-based dashboard, the system addresses key limitations of traditional meters by offering enhanced user interaction and automation capabilities. Key features include slab-based billing for domestic and commercial users, Twilio-enabled SMS alerts for critical conditions, neighbour-aware outage detection using a dual-meter comparison strategy, and a QR code-based UPI payment interface. The system was validated under simulated loads, with real-time data visualized and alerts delivered reliably across platforms. An important enhancement introduced in this work is the storage and visualization of historical data. Energy consumption logs are persistently saved in a structured format (CSV), allowing users to access and analyze past usage trends via dedicated dashboard charts and tables. This capability not only improves billing transparency but also lays the groundwork for future analytics and demand-side energy optimization.

Future work will focus on integrating smart relays for automatic load control during overuse. It will also improve system security with MQTT over TLS and encrypted token-based authentication. We will incorporate machine learning models to predict energy usage, detect anomalies, and optimize billing. Further development will include a scalable multi-user framework with isolated dashboards. We will also implement cloud-based data archiving to support long-term analytics and access across devices.

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